

Education

BITS Goa — M.Sc. Mathematics + B.E. Electronics & Instrumentation Aug 2022 – May 2027

Experience

Data Science Intern — ThinkDigits April 2026 – Present

- Built an end-to-end CV pipeline using **OpenCV**, **scikit-image**, and **NetworkX** for automotive wire harness parsing, achieving **90% wire count accuracy** and **97% noise rejection**.
- Developed a topology-aware wire segmentation system using **Union-Find merging**, **PCA endpoint extraction**, and morphological analysis for electrical rule-compliant routing.
- Built a structured **JSON connectivity graph generator** with dual OCR backends (**PaddleOCR**, **EasyOCR**) for downstream circuit validation.

Summer Intern — Quidich Innovations Lab Jun 2025 – Jul 2025

- Deployed a **production-grade computer vision pipeline** for live ICC broadcast, achieving **0.2 cm terrain precision** via **LiDAR–OpenCV–Open3D integration**.
- Developed a strip-wise intrinsic image decomposition pipeline (**albedo/shading separation**) for high-resolution pitch frames, enabling material-accurate surface analysis for broadcast-grade visual overlays.

Undergraduate Researcher — Sense Lab Jan 2024 – Dec 2024

- Published at **EWSN 2025**: built **Navigator**, an **ESP32-S3** TinyML smartglass using camera–ToF sensor fusion for real-time indoor navigation assistance.
- Trained and quantized on-device **FOMO** and **MobileNetV2** models, achieving **95% TPR**, zero false positives, and sub-**300 ms** inference latency.
- Reduced power draw by **36%** via adaptive frame-rate control, duty-cycled Wi-Fi, and on-device inference, enabling **12+ hour battery life**.

ML Intern — Cilans Systems May 2024 – Jul 2024

- Studied and implemented the **NLA-SLR** research architecture for sign language recognition using **TensorFlow** in a Linux-based training environment.

Projects

Neuro-Symbolic Motion-to-Art Interpreter Jan 2025 – Apr 2025

- Built an end-to-end **AIoT pipeline** combining wearable IMU sensing, embedded DSP, and **LLM-driven motion interpretation** for real-time abstract art generation.
- Engineered a 9-stage **DSP pipeline in C++** for motion-feature extraction, reducing sensor noise by **70%** while sustaining stable **20 Hz** embedded capture.
- Integrated **Google Gemini** and **Stability AI** for neuro-symbolic prompt synthesis and image generation with **fault-tolerant fallback logic**.

Autonomous RL Trading System — PPO + RAG Governance Jan 2026 – Feb 2026

- Built a custom **Gymnasium**-based trading environment modelling portfolio dynamics, exposure-based actions, and transaction cost constraints.
- Trained a **PPO agent (Stable-Baselines3)** over **1M+ timesteps** using technical indicators including **RSI**, **MACD**, momentum, and volatility features.
- Designed a multi-objective reward function achieving **1.5× Sharpe ratio improvement over SPY** and **18% lower maximum drawdown**.

Open Source Contributions

sktime - European Summer Of Code

- Contributed to **sktime** by implementing the **HampelFilter** detector, optimizing **RelativeLoss** evaluation throughput, and improving QA coverage for **MovingWindow** and **ClaSPTransformer**.

Leadership Experience

Team Lead, Perception & Navigation — Project Kratos (Mars Rover) Sept 2023 – Apr 2025

- Led perception, navigation, and arm, life detection subsystems development for a **50+ member** ROS-based autonomous rover team.
- Developed closed-loop rover autonomy using **YOLOv5**, **CUDA-accelerated point-cloud mapping**, trajectory-aware planning, and **RTK-GPS fusion**, achieving **3 cm localization precision**.
- Engineered a **vision-based gesture override system** enabling human-in-the-loop recovery during autonomous rover operations.

Skills

Languages/Systems: Python, C++, C, Java, SQL, Docker, ESP32, ESP-IDF, DSP, Edge AI

ML/CV/Robotics: PPO, PyTorch, TensorFlow, scikit-learn, OpenCV, ROS, CUDA, LiDAR, Open3D, RAG, FAISS, Sensor Fusion